

RES-5 higher-loop common-mode bound: the residual is the 2PI skeleton

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The framing correction (res5-oneloop-loop-disentangling) located RES-5 as the loop expansion beyond the exact one-loop. The operator directed bounding the higher-loop DIFFERENCE $|\sum_{\ell \geq 3} \Delta F_{\ell}^{\text{pd}}| < \Delta F_{\text{margin}}$. This note frames that residual correctly – as the 2PI SKELETON expansion – and records the favourable leading datum. It does NOT close the bound. Scope: structural framing + the base skeleton; anchor μ^2 .

2. The residual is the 2PI skeleton expansion, not bare loops

The dressing

$$\hat{r}(I) = r_R + 2\lambda' I$$

is the SELF-CONSISTENT Hartree resummation: the leading (tadpole/Hartree) loops are already summed into $\hat{r}$. Therefore the one-loop log-det with $\hat{r}$ is the resummed mean field, and the genuine higher-loop corrections are the 2PI SKELETON diagrams built from the DRESSED propagator $G_d = (\hat{r} + c(q^2 - q_0^2)^2)^{-1}$ – not bare loops. This is why the bare-loop / bare-susceptibility counting was the wrong object (it double-counted the already-resummed dressing). The skeleton expansion has softer (dressed) propagators and is better-behaved.

3. The leading skeleton (sunset) is favourable

The leading 2PI skeleton beyond Hartree–Fock is the SUNSET (2-loop), which is exactly SC-SCOPE’s third cumulant. At the DIFFERENCE level (common-mode subtracted), SC-SCOPE found it thin:

$$s_3 = \frac{|\Delta F_{\text{sunset}}^{\text{pd}}|}{\Delta F_{\text{margin}}} \sim 4\% \ll \frac{1}{2} \text{ (the geometric convergence radius).}$$

The base skeleton sits a factor ~ 12 inside the convergence region – the strong common-mode part has cancelled, leaving only the a_0 -suppressed pattern-dependent sunset.

4. The closure structure and the residual

With the common-mode cancellation $\Delta F_\ell^{\text{cm}}[\mathcal{P}] - \Delta F_\ell^{\text{cm}}[\mathcal{R}_H] = 0$ at every loop order ℓ , the matched-order-to-exact margin survives iff the higher-skeleton DIFFERENCE is geometrically dominated:

$$\left| \sum_{\ell \geq 3} \Delta F_\ell^{\text{pd}} \right| \leq \Delta F_{\text{margin}} \sum_{\ell \geq 3} s_\ell < \Delta F_{\text{margin}} \quad \text{if } s_\ell \leq s < \frac{1}{2} \quad \forall \ell.$$

The base $s_3 \sim 4\%$ is favourable; the residual is the all-order skeleton domination $s_\ell \leq s < \frac{1}{2}$. **This is the genuine RES-5 frontier: a dedicated 2PI skeleton computation of the higher-loop DIFFERENCE** – bounding the dressed three-loop and higher skeletons after common-mode subtraction. It is not closeable by the elementary means of this session.

5. Status

T3 PROOF SKETCH. The residual is correctly framed as the 2PI skeleton expansion (self-consistent dressing), with the sunset (SC-SCOPE, $s_3 \sim 4\%$) as the favourable leading skeleton, well inside the convergence radius. **OPEN (the frontier):** the all-order skeleton domination $s_\ell < \frac{1}{2}$ (the dressed higher-skeleton DIFFERENCE bound) – a dedicated 2PI computation. No tier flip: B1-RH-ENUM T6 CONDITIONAL on {H-LAYER}. **B1 closure** $\iff$ the higher-skeleton difference bound.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “Another reframing, still no bound.” VALID: this note corrects the OBJECT (2PI skeleton, not bare loops) and records the favourable base, but does not bound the higher skeletons – that is the dedicated 2PI residual, honestly marked. The reframing is necessary (the bare-loop object was wrong).
- (β) “ $s_3 \sim 4\%$ is one datum; the higher s_ℓ could grow.” VALID and UNRESOLVED: the base is favourable, but the all-order skeleton domination is the residual. The dressed propagators (softer than bare) and the common-mode $+a_0$ suppression make it plausible, not proved.
- (γ) “Does this change B1?” DISMISSED: it sharpens the residual map (2PI skeleton), not any tier. B1 stays T6 on {H-LAYER}.

Quantitative sanity check (mandatory): *dressing* – $\hat{r} = r_R + 2\lambda'I = 0.305 + 0.032 = 0.337$ (Hartree-resummed); *base* – $s_3 \sim 4\% \ll \frac{1}{2}$ (factor ~ 12 inside); *structure* – common-mode cancels per loop order, leaving the a_0 -suppressed pattern-dependent skeleton; *consistency* – the sunset is the leading 2PI skeleton, matching SC-SCOPE’s two-loop datum.

7. Result footer

Result ID:	B1-RH-ENUM / RES-5 higher-loop common-mode (2PI skeleton)
Precise statement:	The dressing $r_{\text{hat}} = r_R + 2 \lambda' I$ is the self-consistent Hartree resummation, so the RES-5 higher-loop residual is the 2PI SKELETON expansion (dressed propagators), not bare loops. The leading skeleton (sunset/2-loop = SC-SCOPE third cumulant) DIFFERENCE is $s_3 \sim 4\% \ll 1/2$. Closure:

	$ \sum_{l \geq 3} \Delta F_l^{\text{pd}} \leq \Delta F_{\text{margin}} \sum s_l < \Delta F_{\text{margin}}$ if $s_l \leq s < 1/2$ for all l . Base favourable; all-order skeleton domination is the residual.
Scope:	Structural framing + base skeleton; anchor μ^2 . The all-order skeleton bound is OPEN (dedicated 2PI).
Dependencies:	res5-oneloop-loop-disentangling (the framing); SC-SCOPE (sunset = leading skeleton, $\sim 4\%$); res5-commonmode-envelope (a0); Math424-AddA.
Evidence grade:	ANALYTIC (framing) + EXECUTED (res5_higherloop_skeleton.py 3/3) + INHERITED (SC-SCOPE s_3).
Reproduction command:	python codes/vacuum/res5_higherloop_skeleton.py
Expected output:	$r_{\text{hat}}=0.337$ (Hartree shift 0.032); $s_3 \sim 4\% < 0.5$; 3/3 PASS.
Falsification gate:	A dressed higher skeleton with $s_l \geq 1/2$ (would break the geometric domination); or a demonstration that the skeleton expansion is not the right object (e.g. a non-skeleton contribution at the difference level).
Tier before / after:	No tier flip. B1 T6 on {H-LAYER}. RES-5 residual = the 2PI higher-skeleton difference bound (base $s_3 \sim 4\%$ favourable).
No-overclaim statement:	This does NOT close RES-5. It frames the residual as the 2PI skeleton expansion and records the favourable leading skeleton (SC-SCOPE $\sim 4\%$); the all-order skeleton domination is the dedicated 2PI frontier. B1 stays T6 on {H-LAYER}.
Next required action:	Compute/bound the dressed three-loop skeleton DIFFERENCE (the next term after the sunset) and establish the skeleton-parameter domination $s_l < 1/2$; this is a dedicated 2PI effective-action computation.